NORTH SOUTH UNIVERSITY

Department of Electrical and Computer Engineering

Course: CSE435L
Course Title: Introduction to VLSI
Section: 01
Report: A semester overview

Shafin Rahman 2233146642

Faculty: Iqbalur Rahman Rokon
Lab Instructor: Sariah Tul Qarim
Submitted to: Sariah Tul Qarim

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NOT Gate:

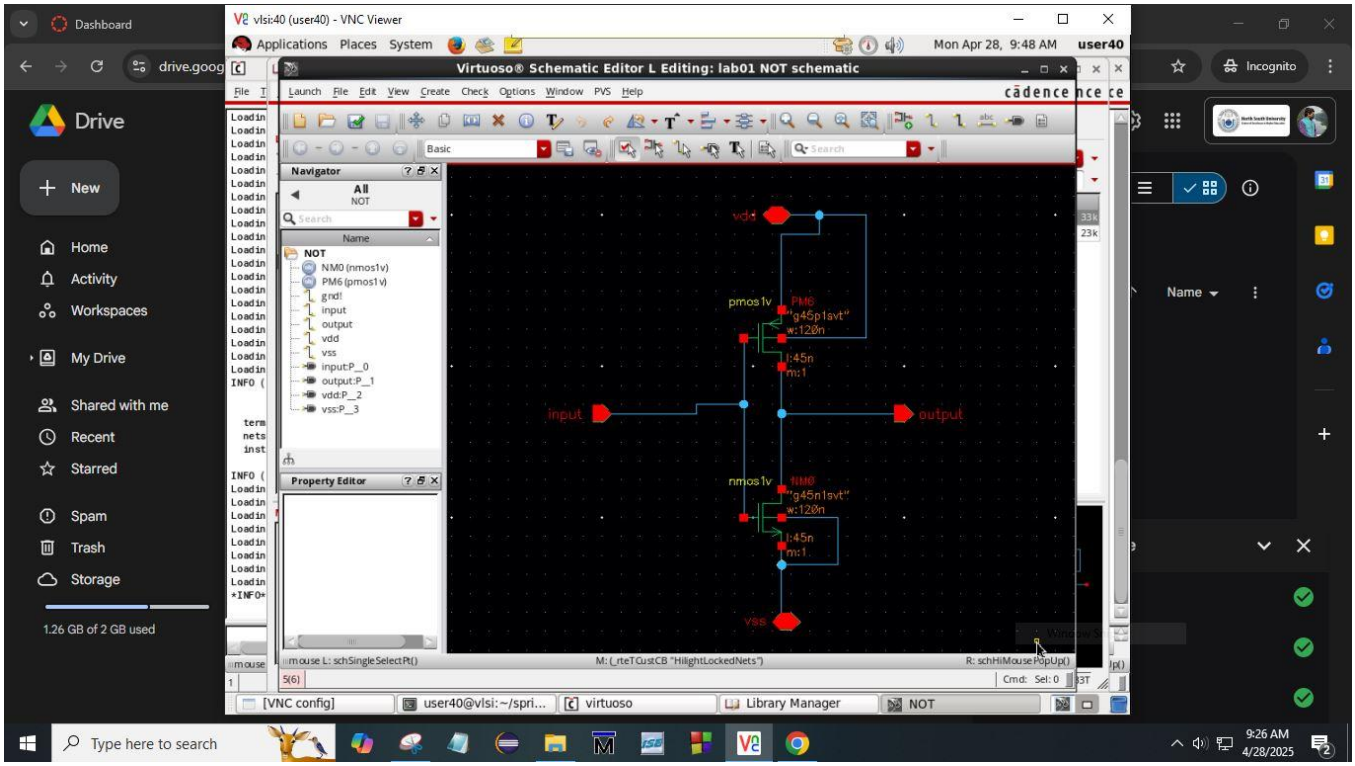


Figure 01: NOT gate without electrical components

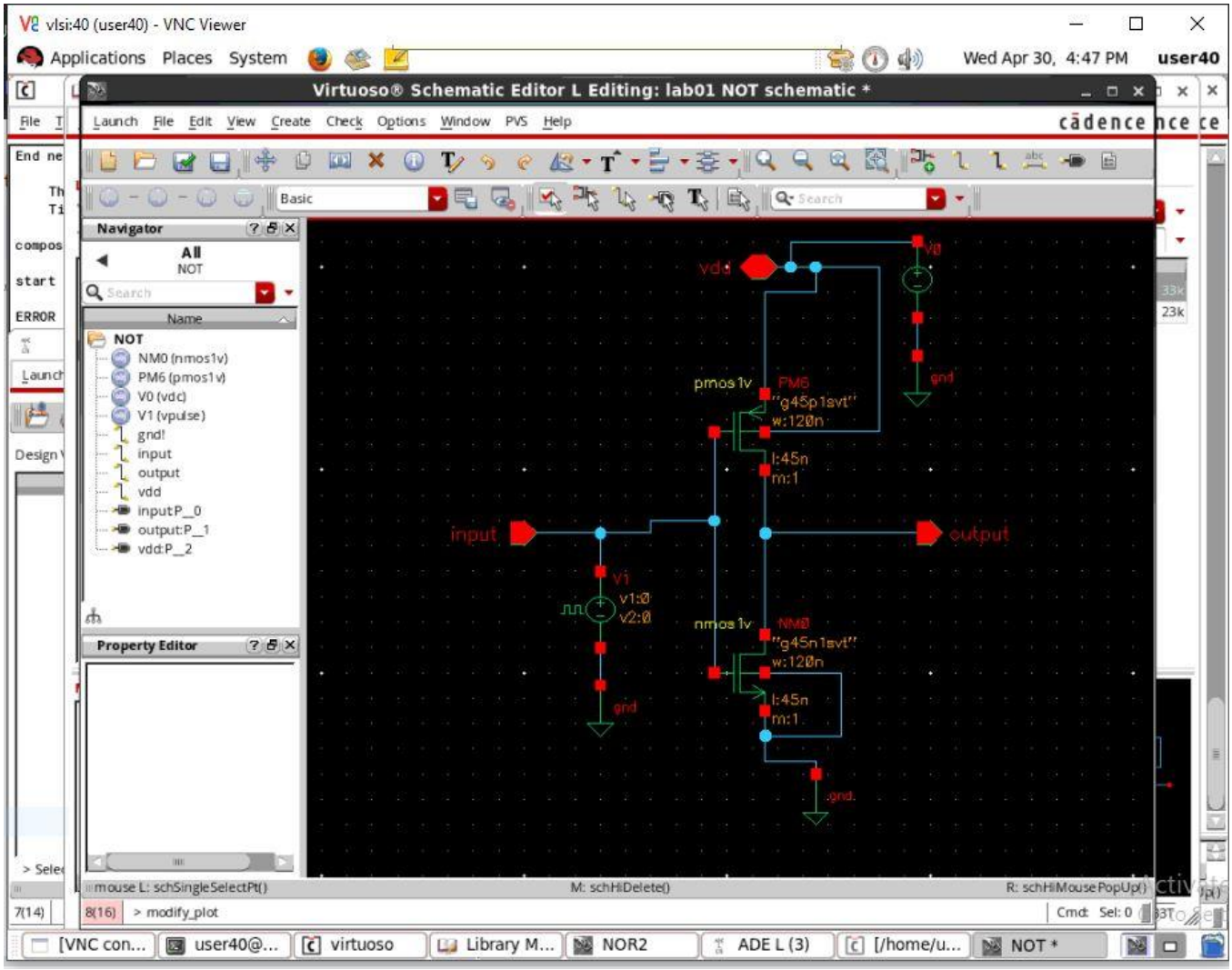


Figure 02: NOT gate connected with electrical components



Figure 03: Transient analysis of NOT gate

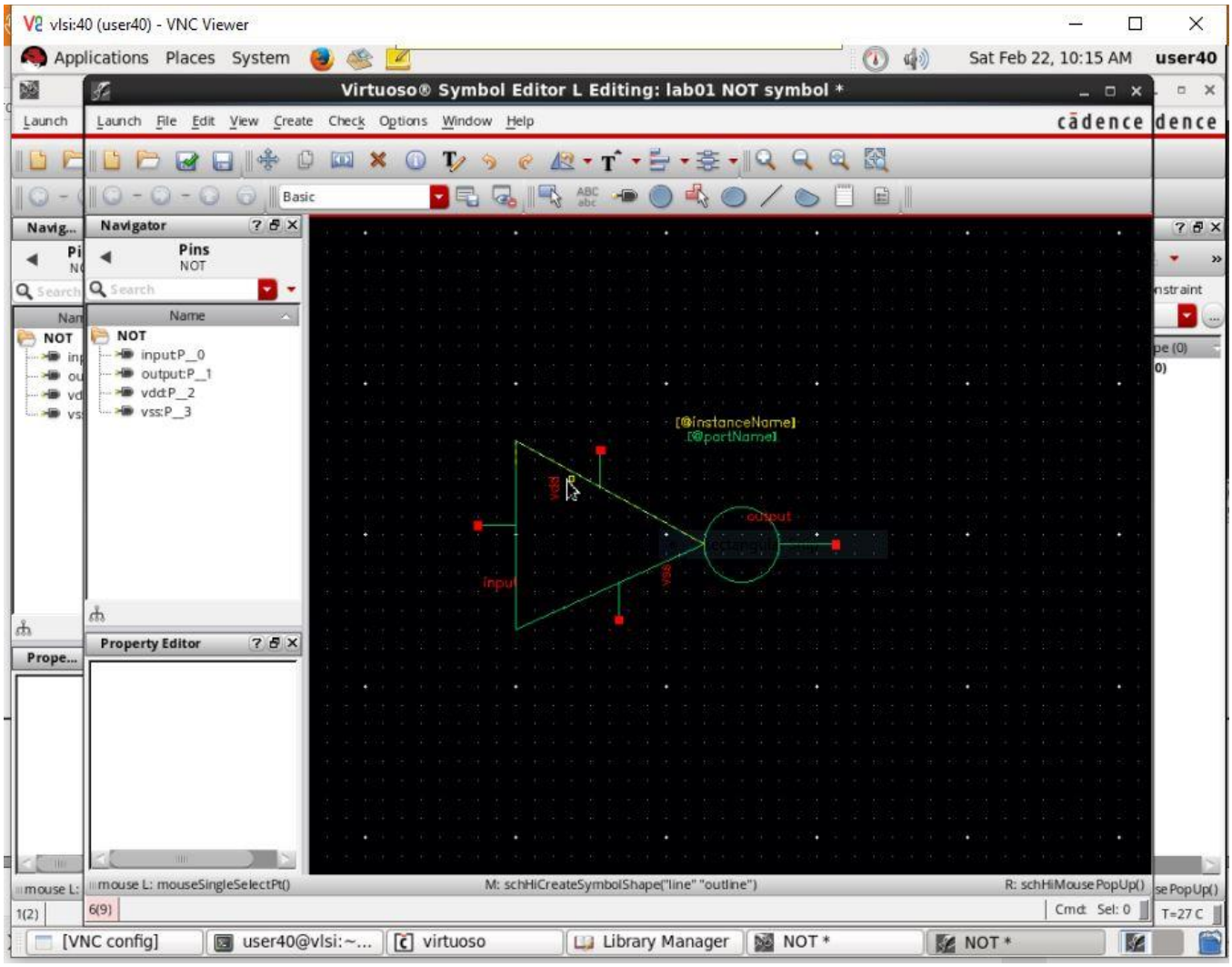


Figure 04: NOT gate symbol

NAND2:

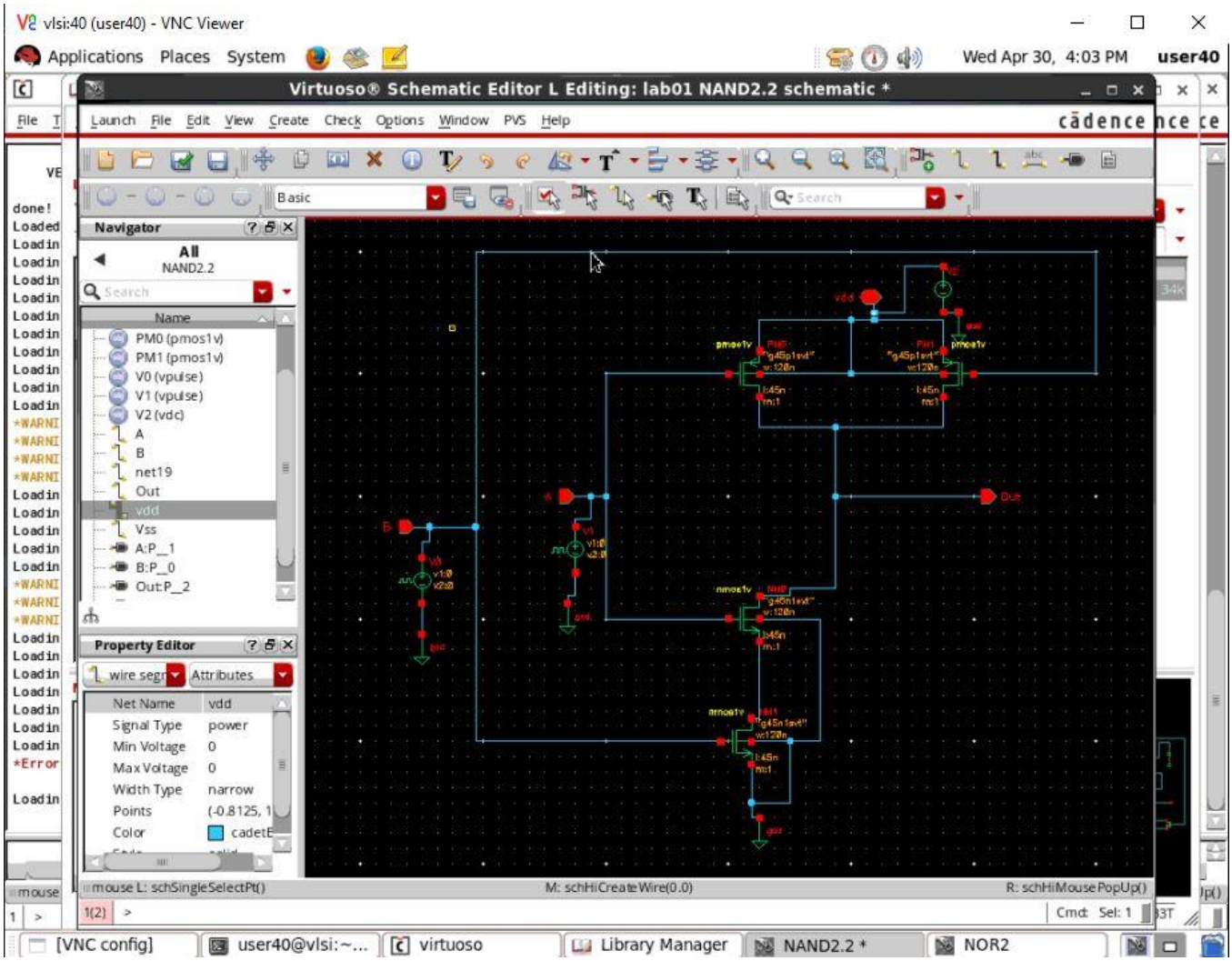


Figure 05: NAND Gate with electrical components

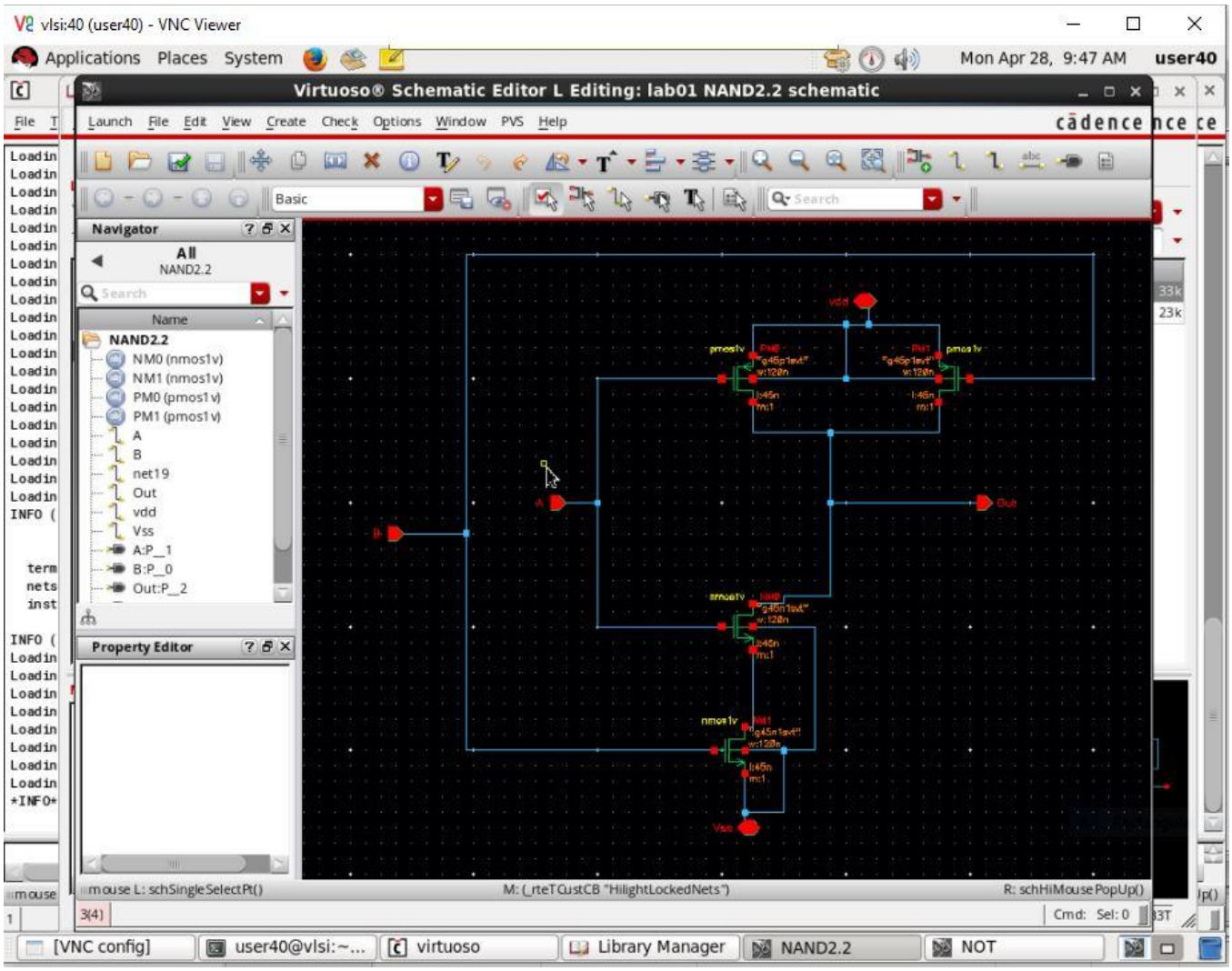


Figure 06: NAND Gate without electrical components

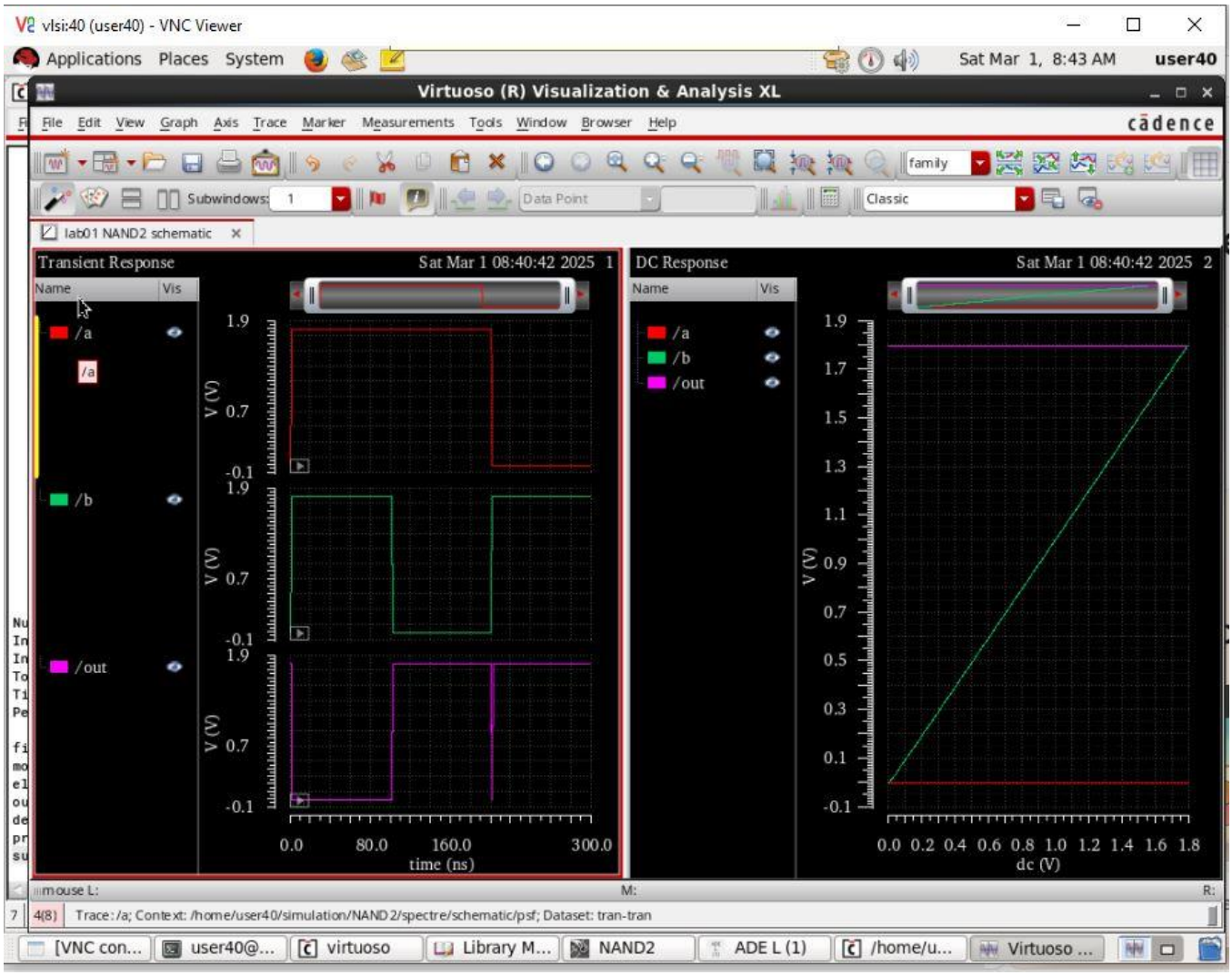


Figure 07: NAND Gate output and Transient Analysis side by side

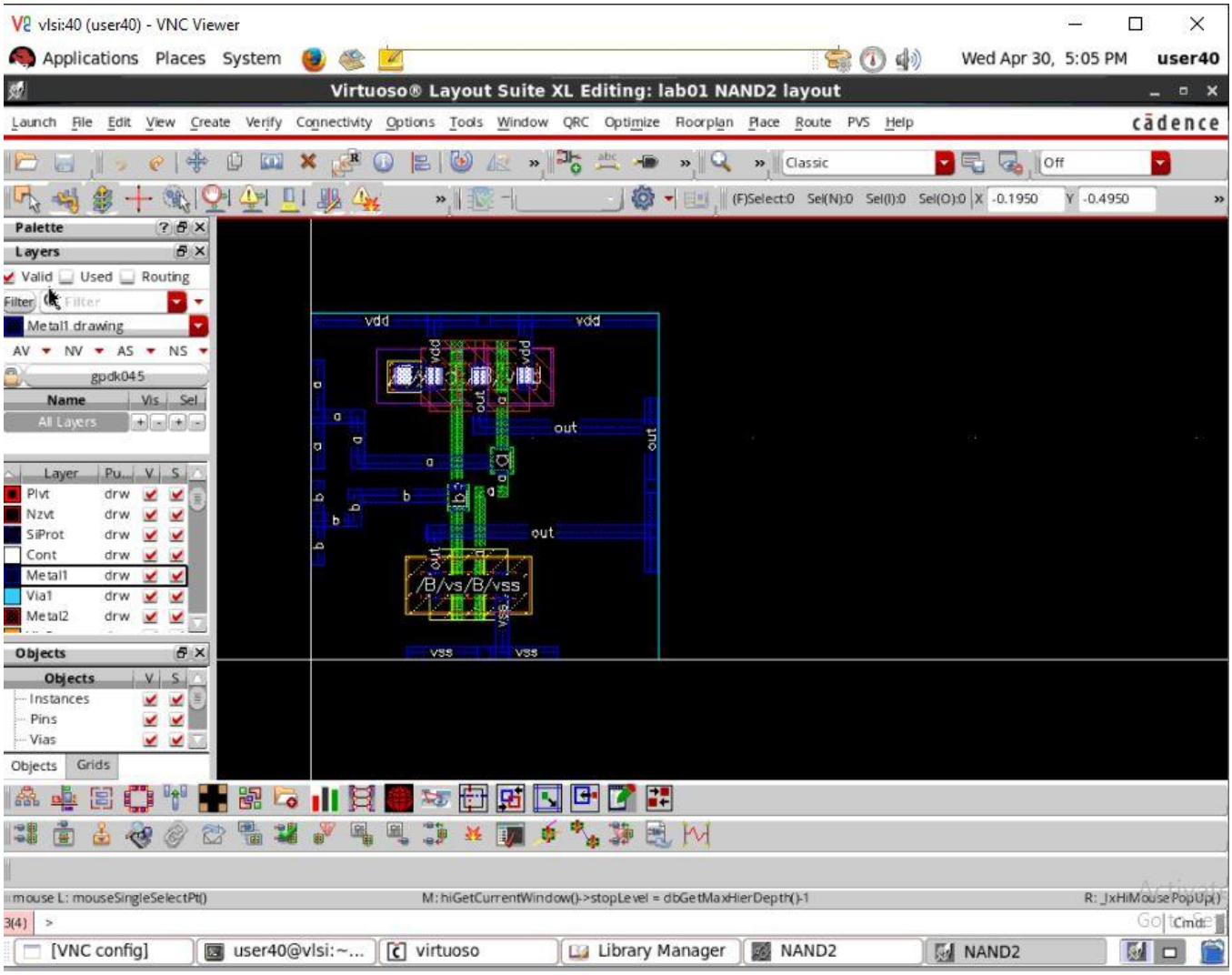


Figure 08: Layout of NAND2

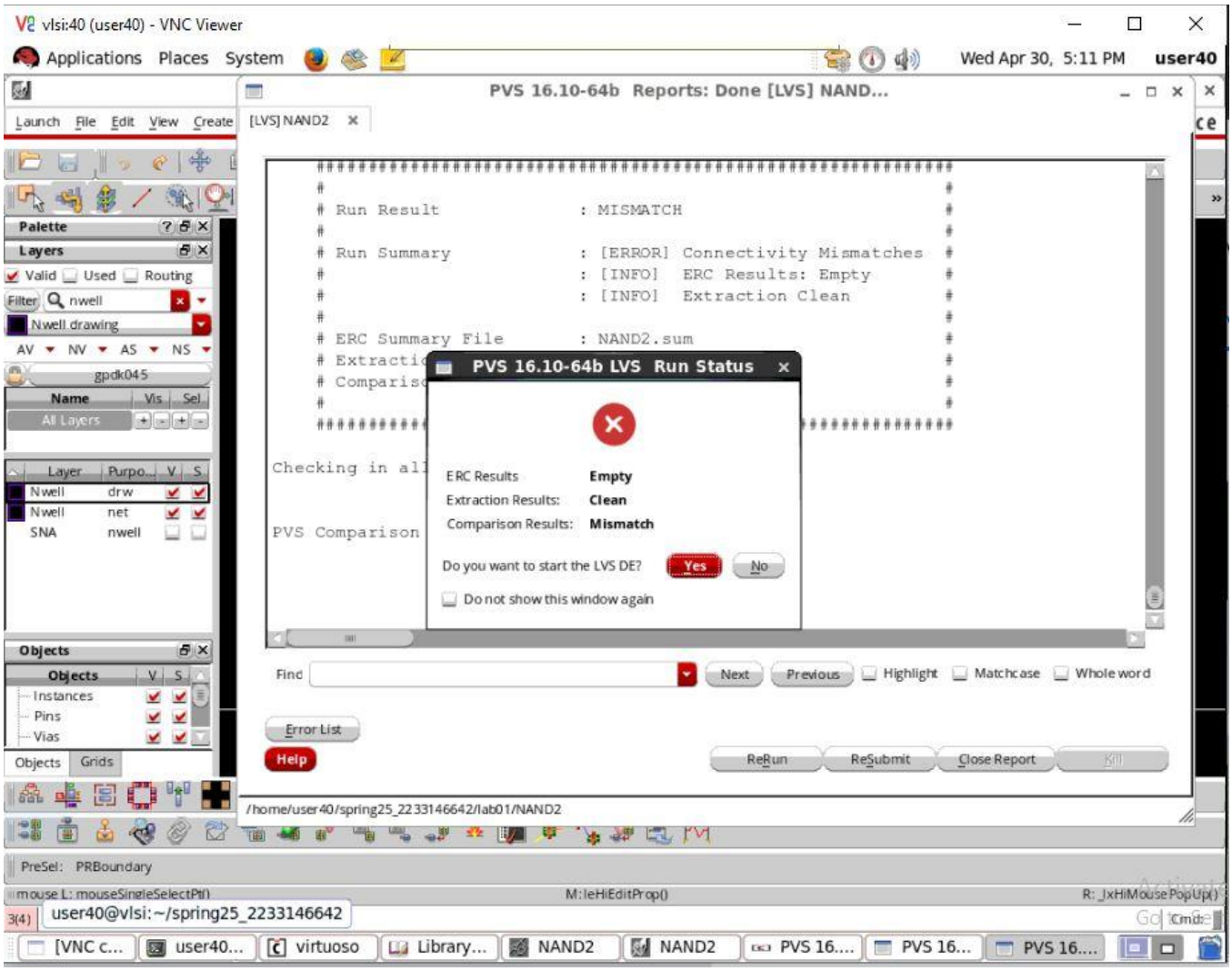


Figure 09: NAND2 LVS run output

NOR2 :

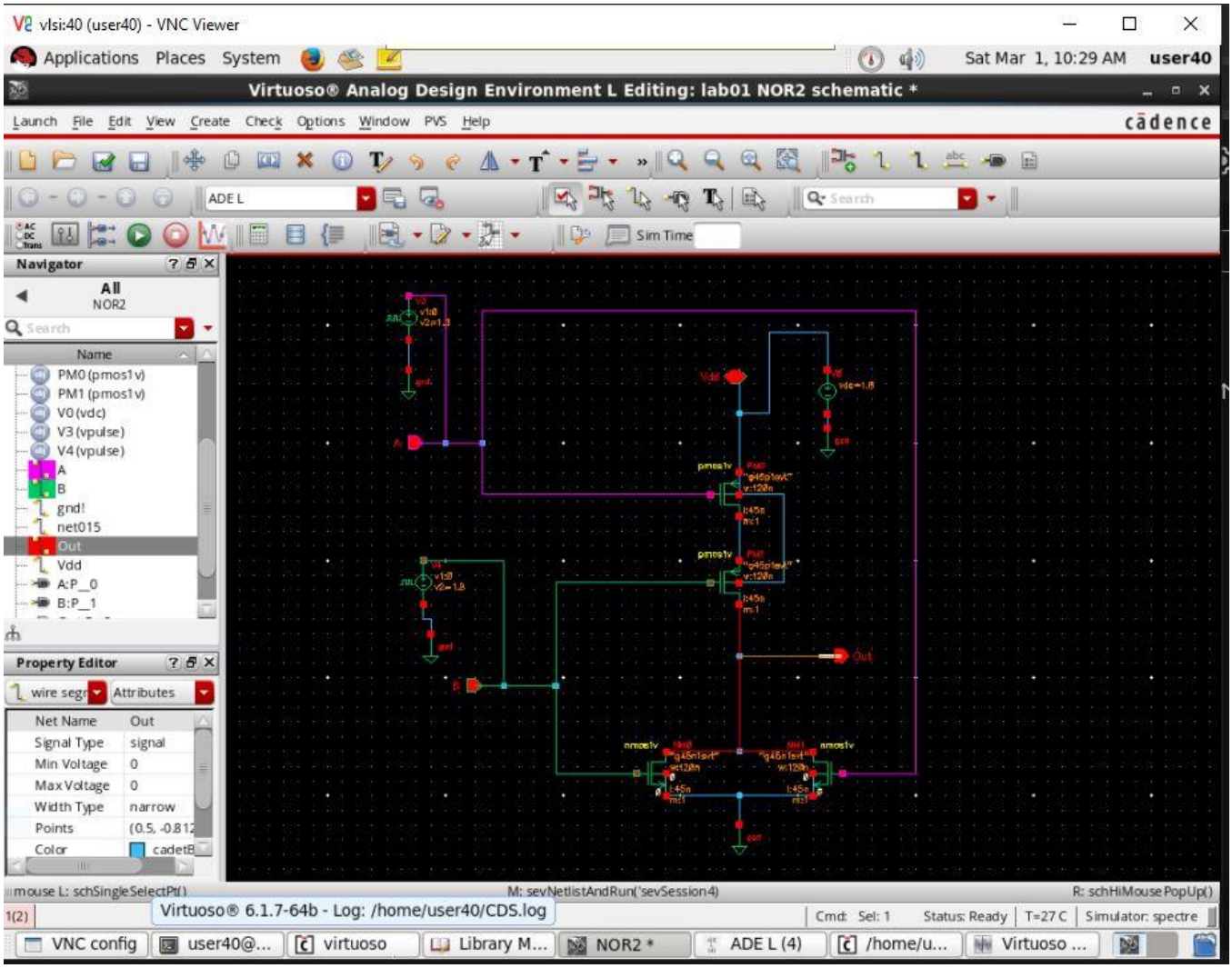


Figure 10: NOR2 schematic with electrical components



Figure 11: NOR2 DC analysis

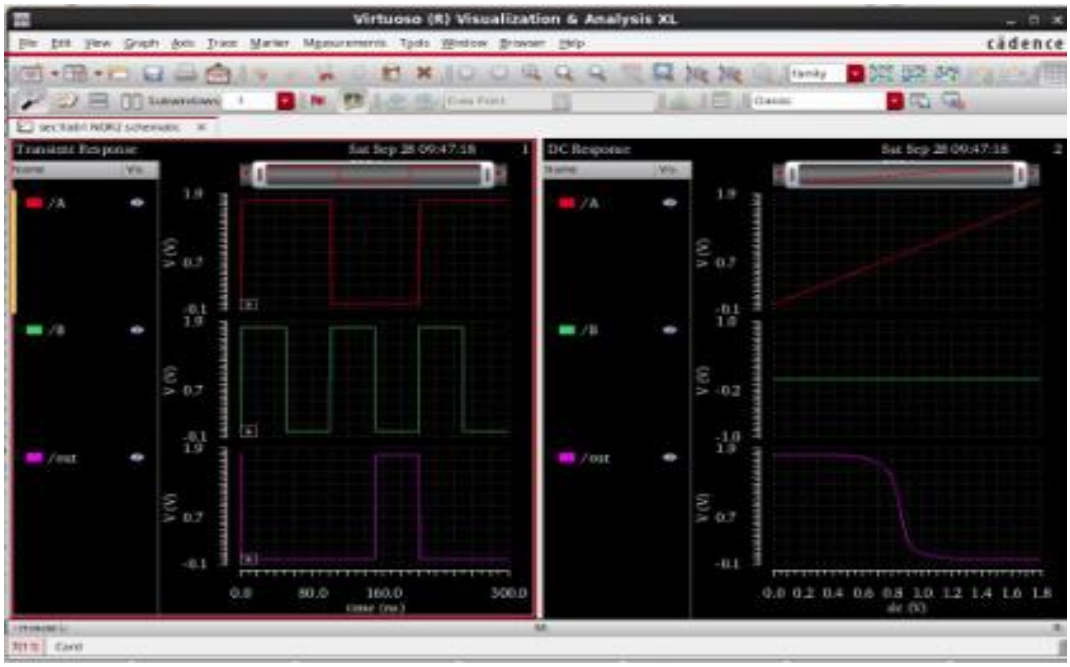


Figure 12: NOR2 DC & Transient analysis

Project (OR2 Gate):

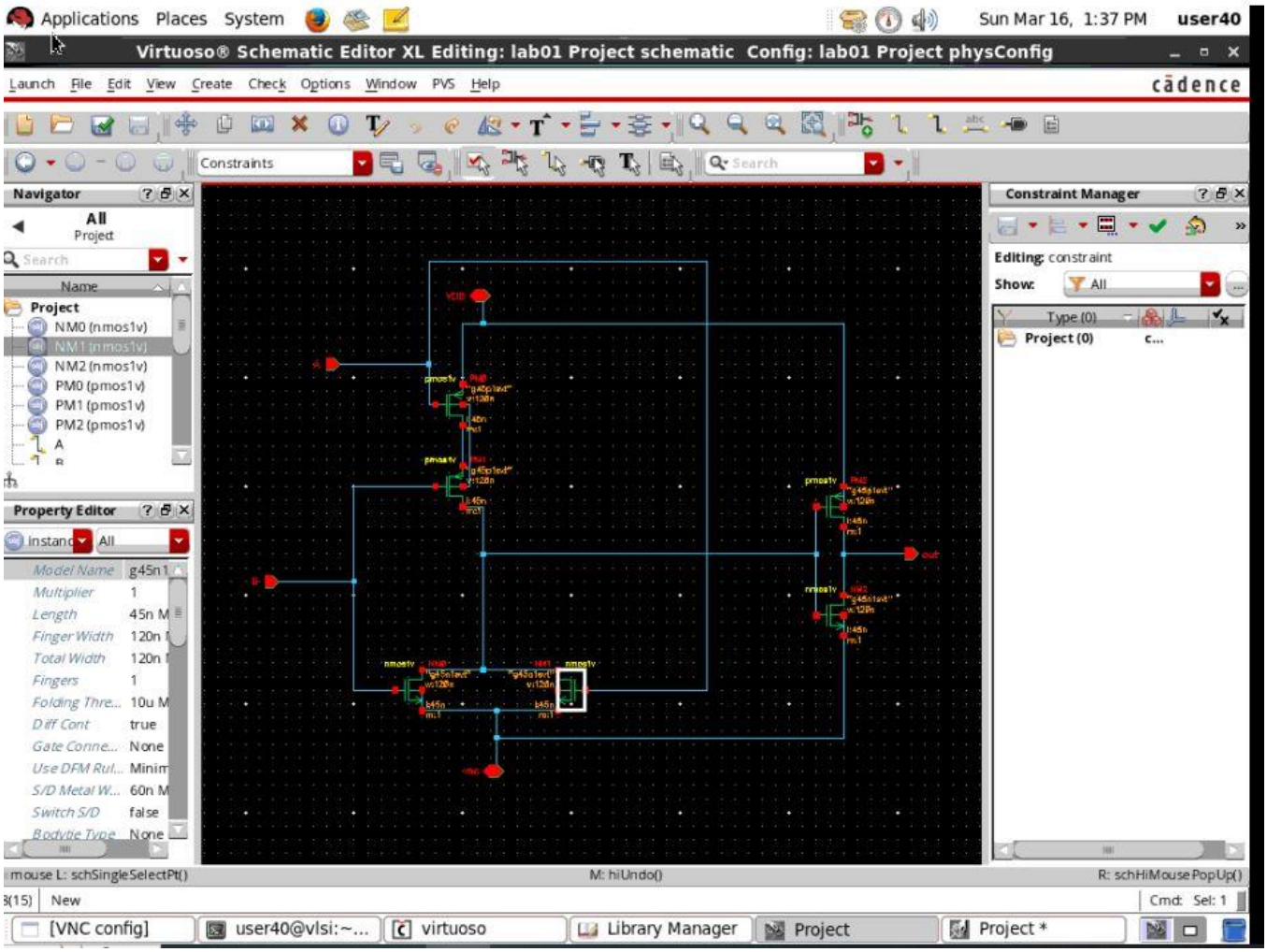


Figure 13: OR2 schematic without electrical components

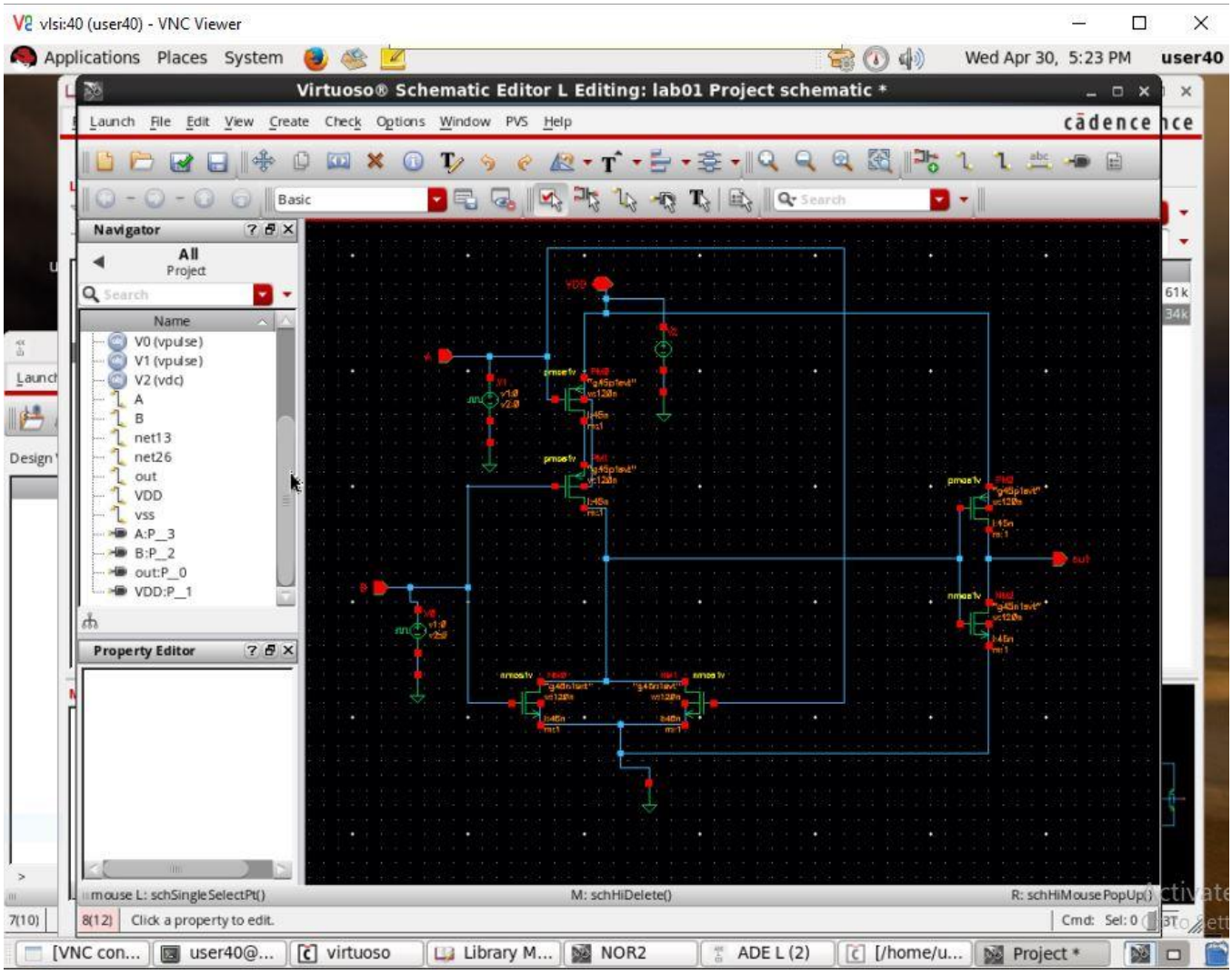


Figure 14: OR2 schematic with electrical components

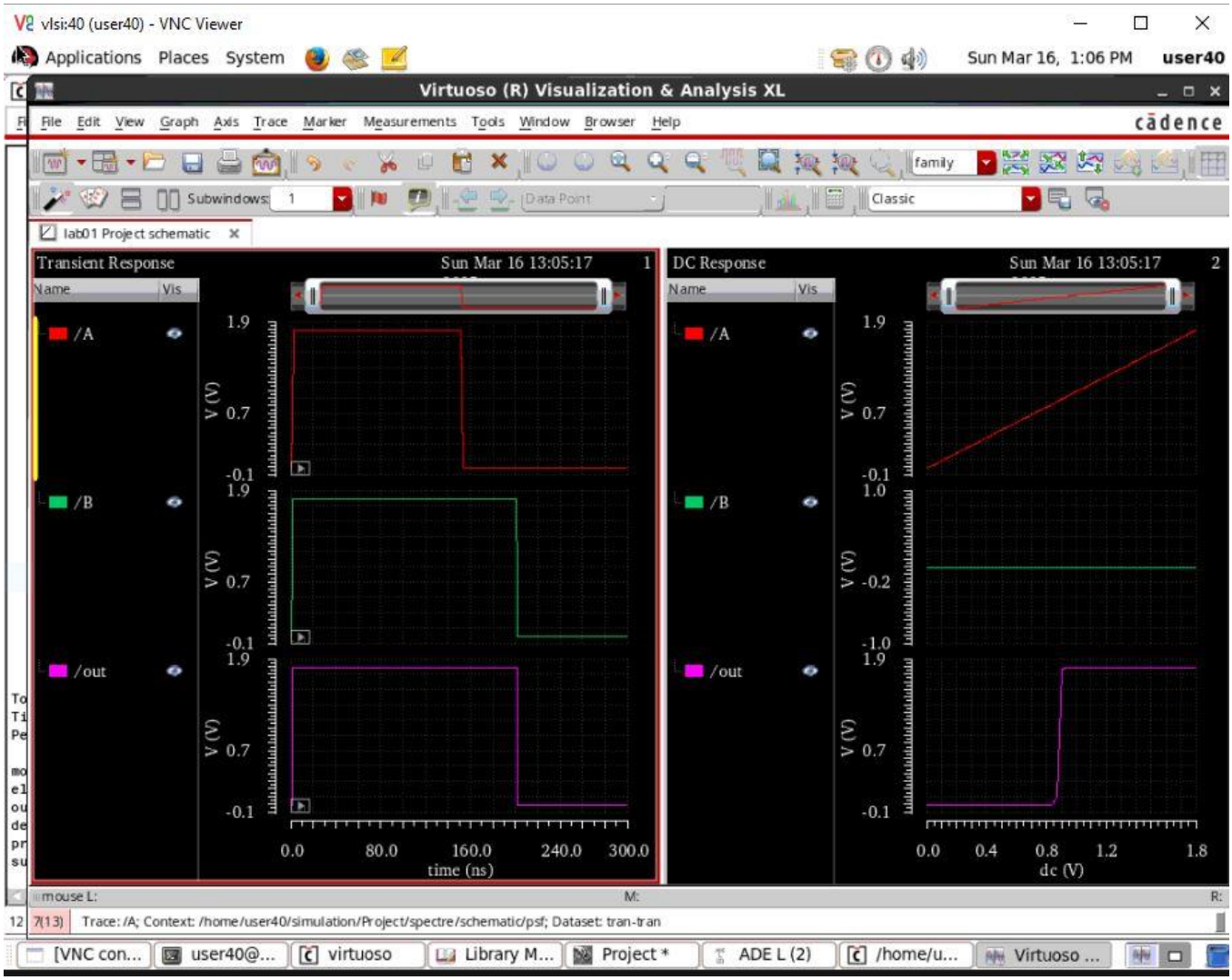


Figure 15: Transient and DC analysis of OR2

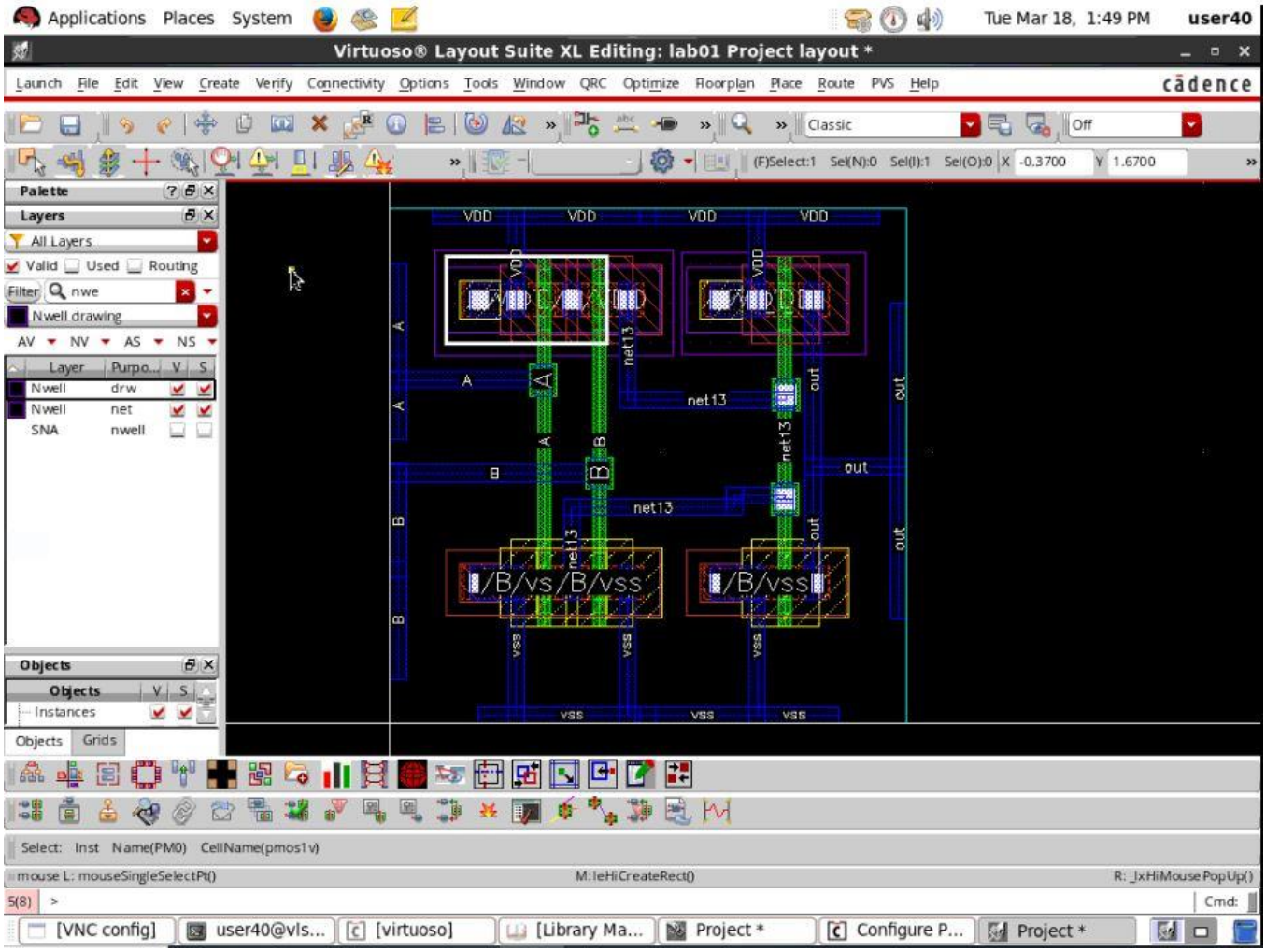


Figure 16: Layout of OR2

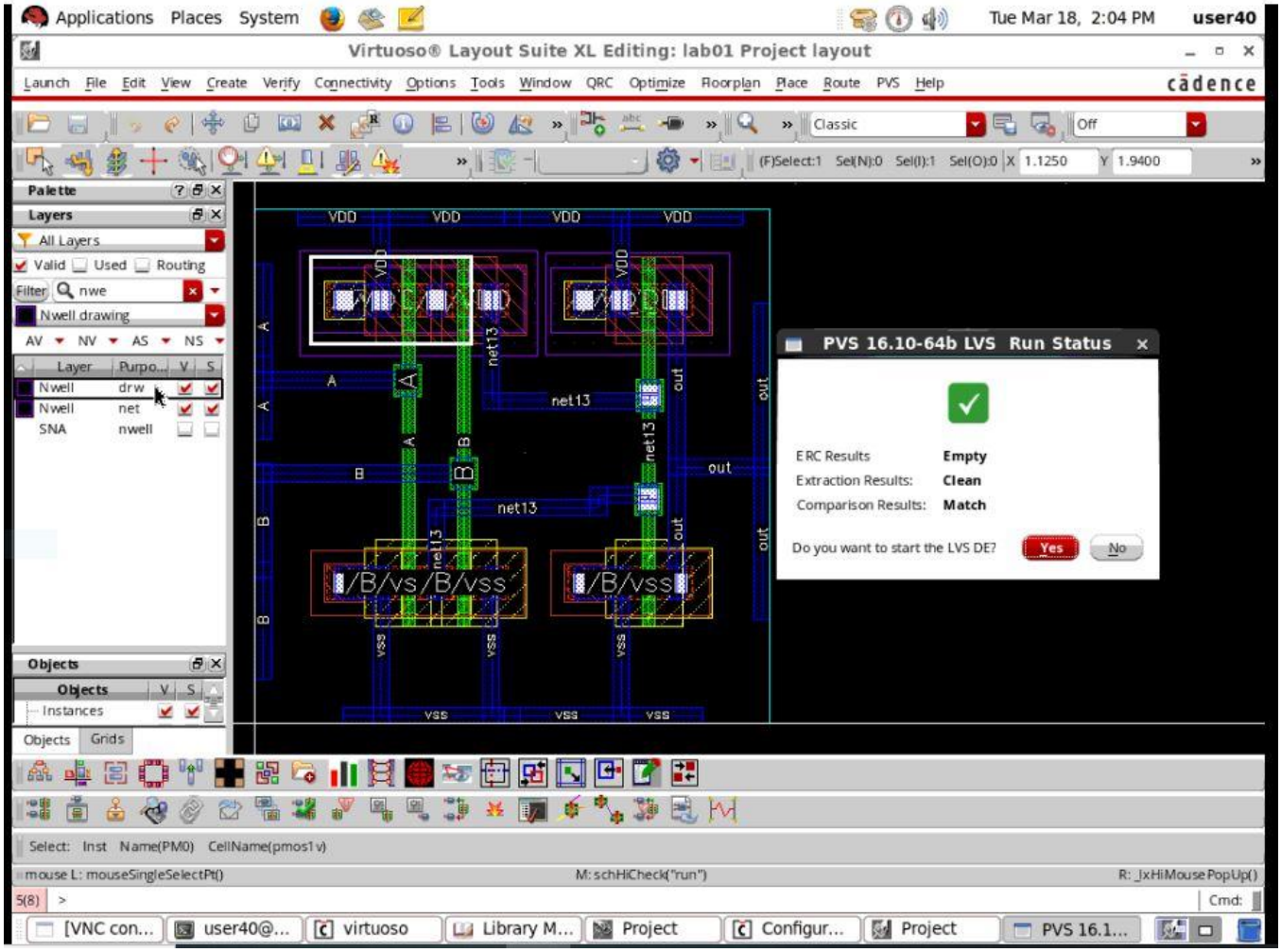


Figure 17: LVS of OR2

Project(MUX 2x1) using NAND2 and NOT:

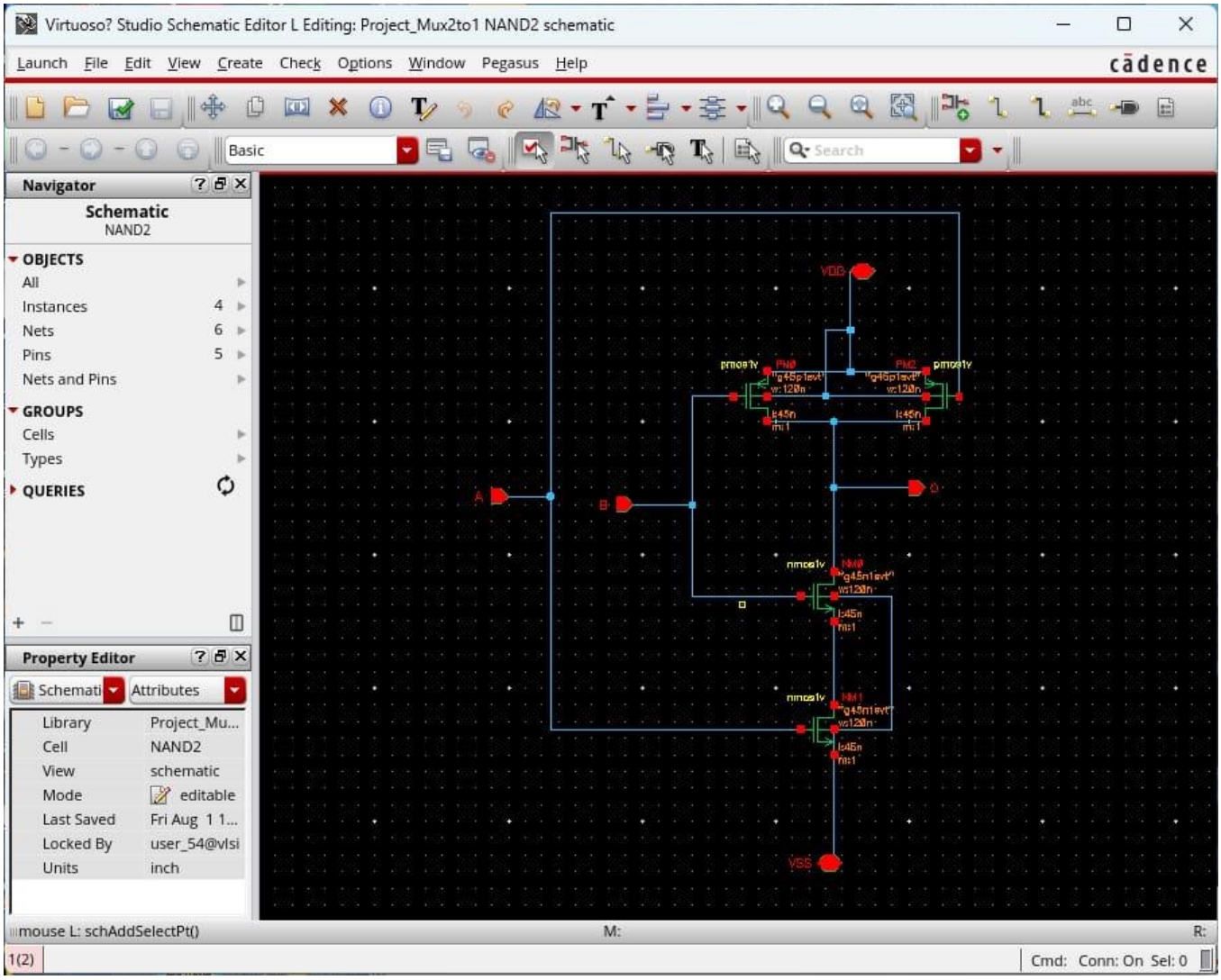


Figure 17: MUX 2x1 NAND2 Schematic

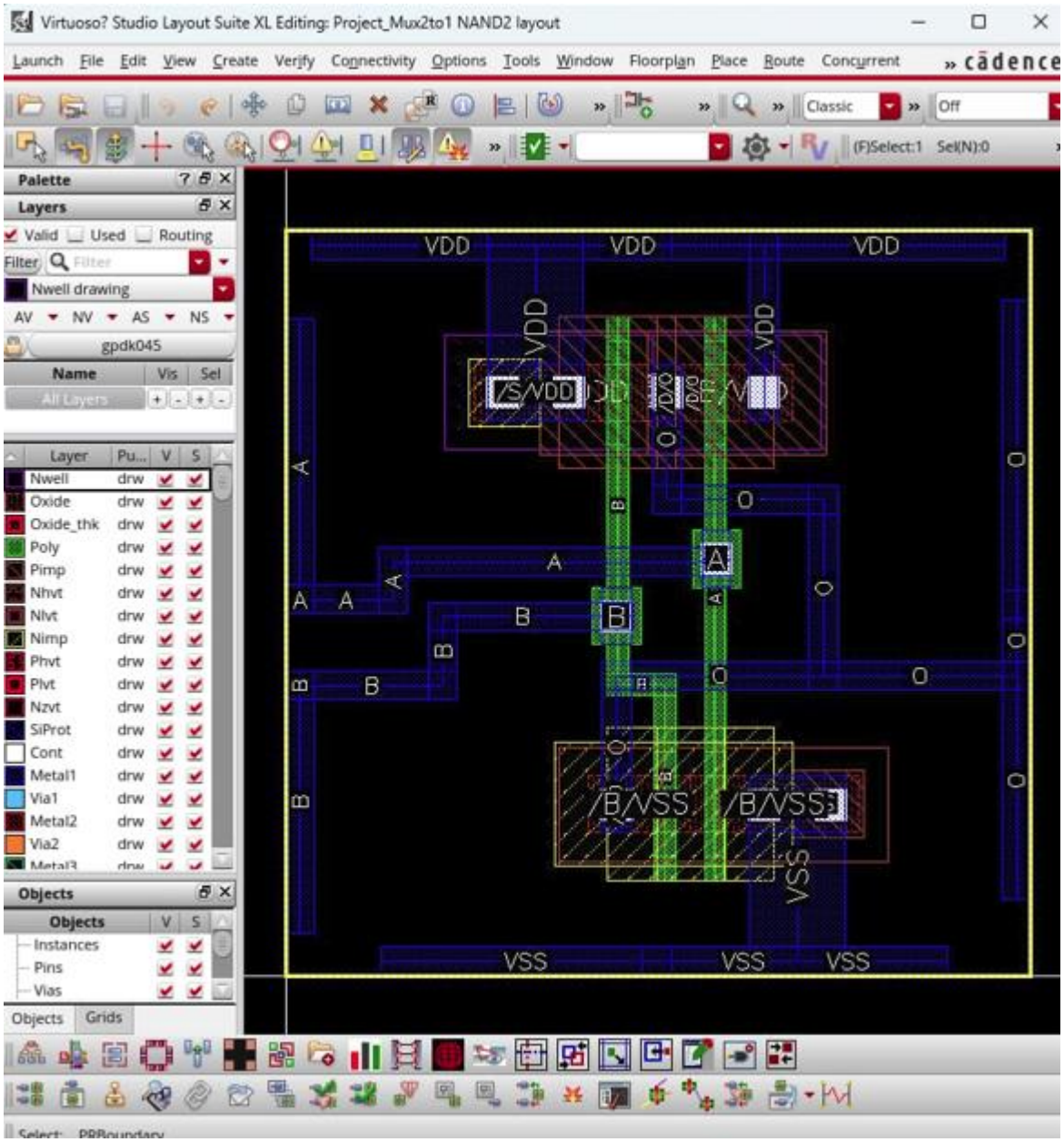


Figure 18: MUX 2x1 NAND2 Layout

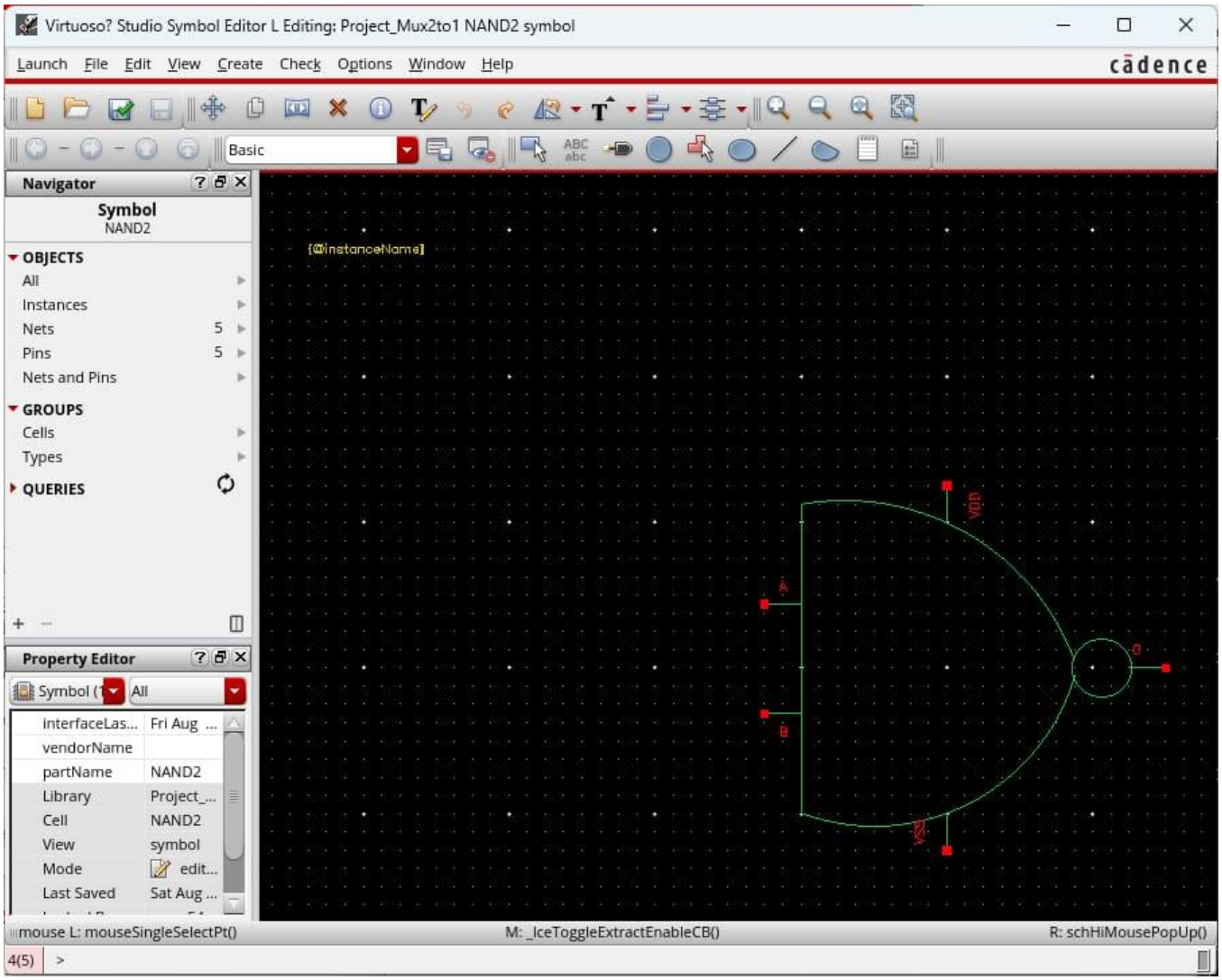


Figure 19: MUX 2x1 NAND2 symbol

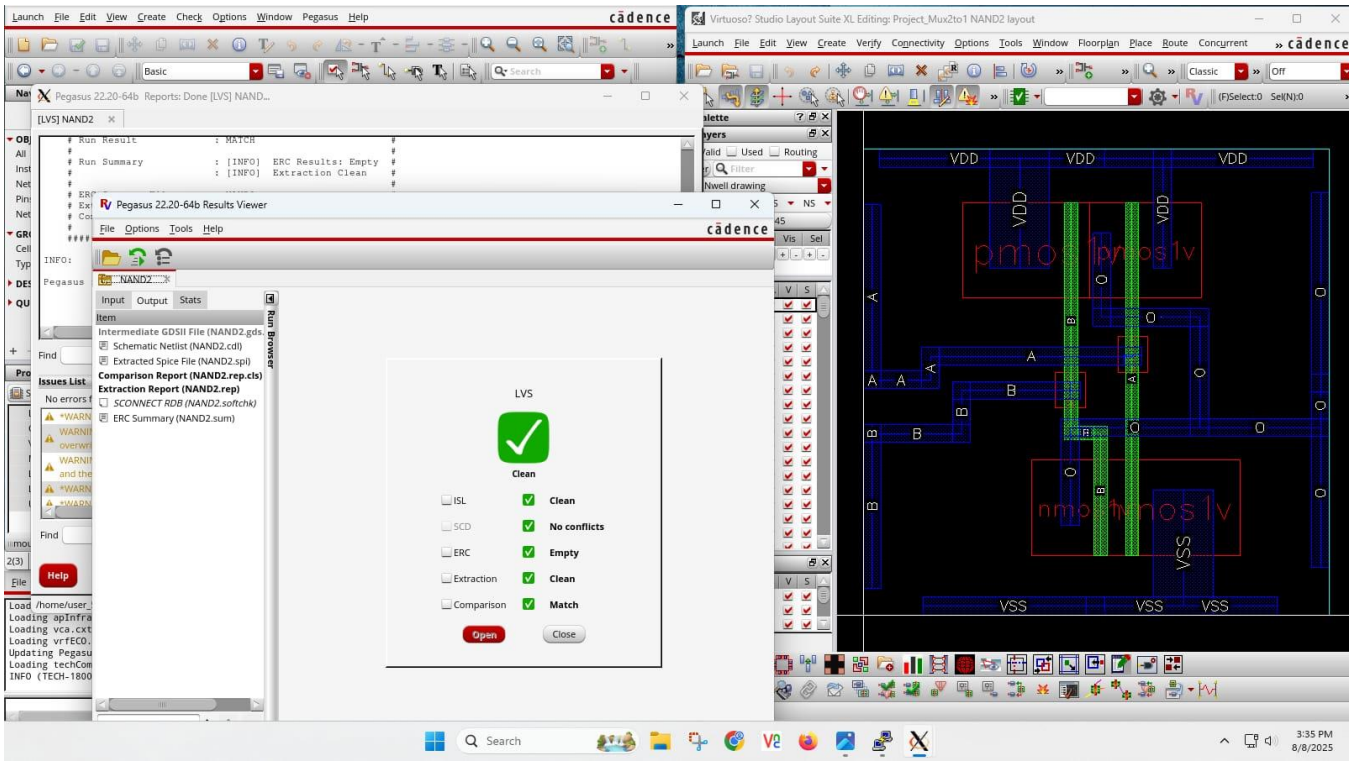


Figure 20: LVS output of NAND2 of MUX 2x1

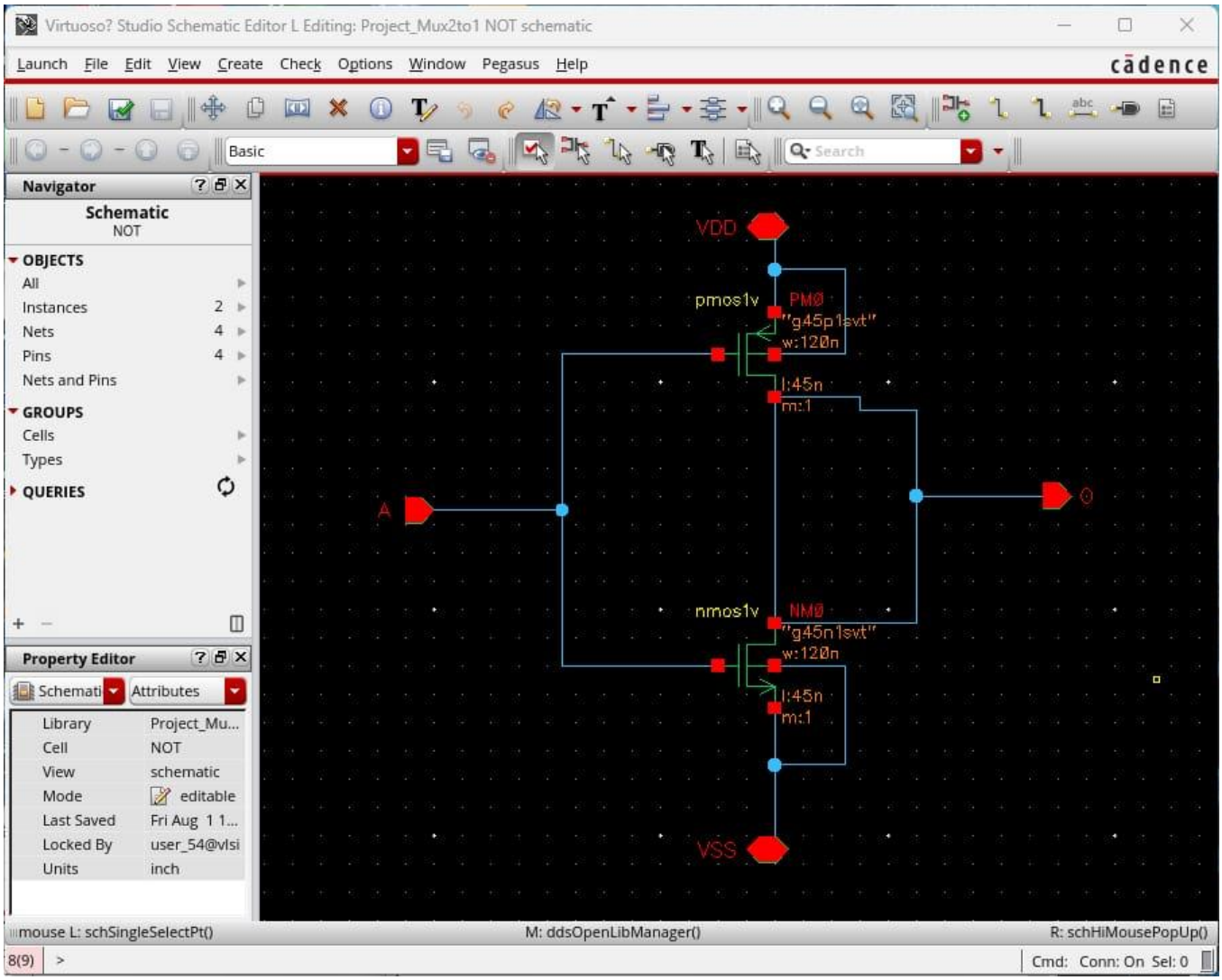


Figure 21: NOT Schematic of MUX 2x1

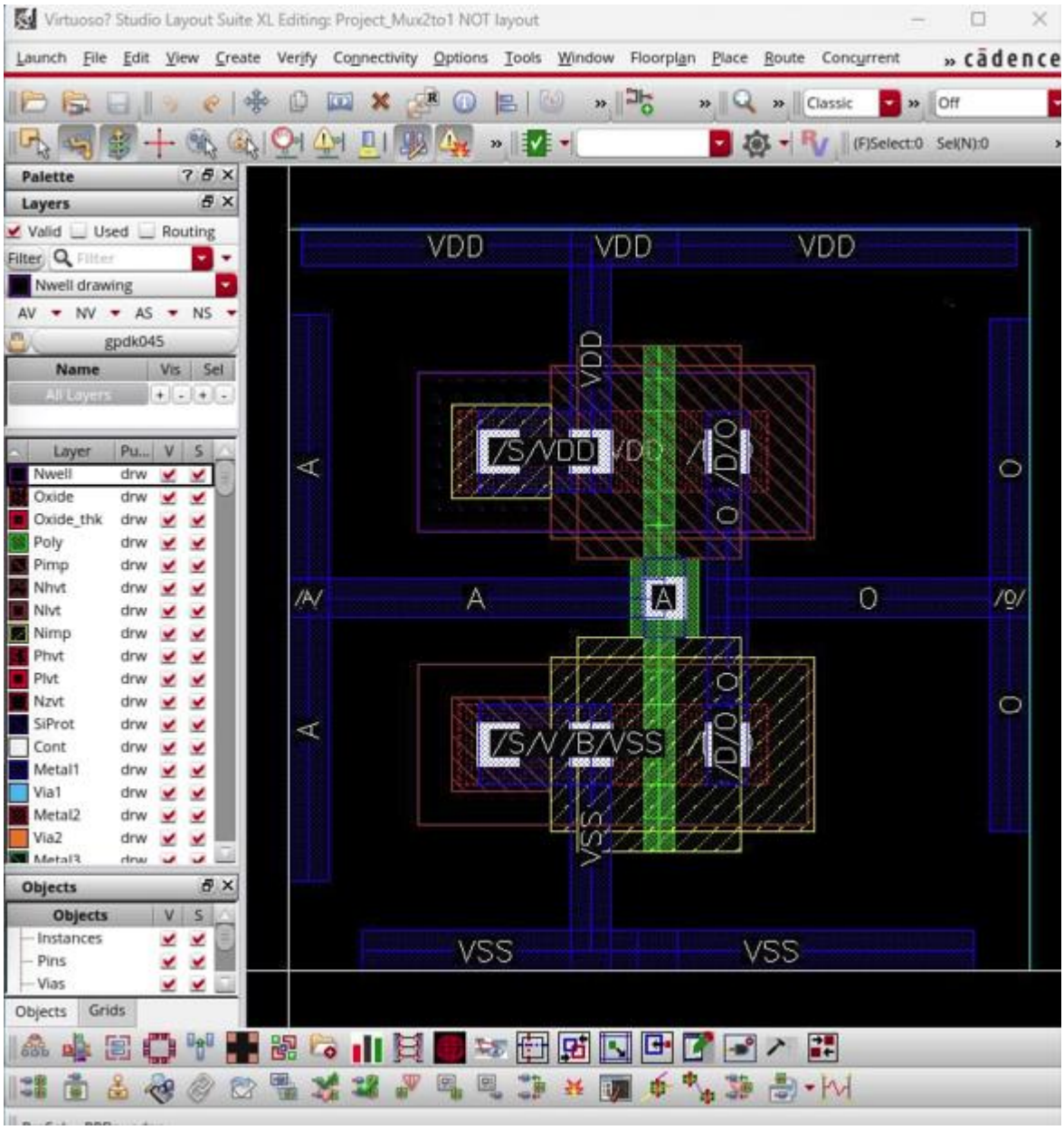


Figure 22: NOT layout of MUX 2x1

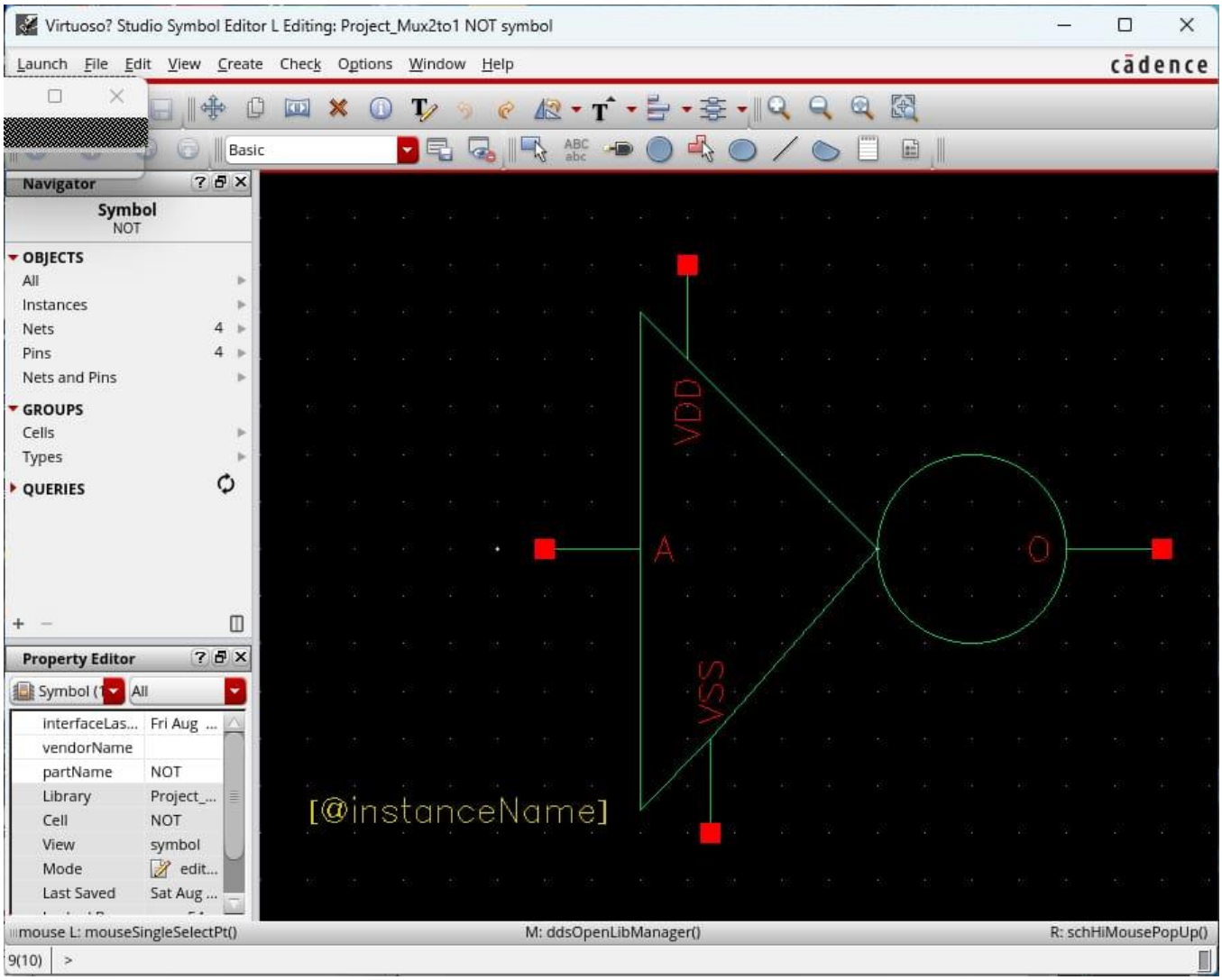


Figure 23: NOT symbol

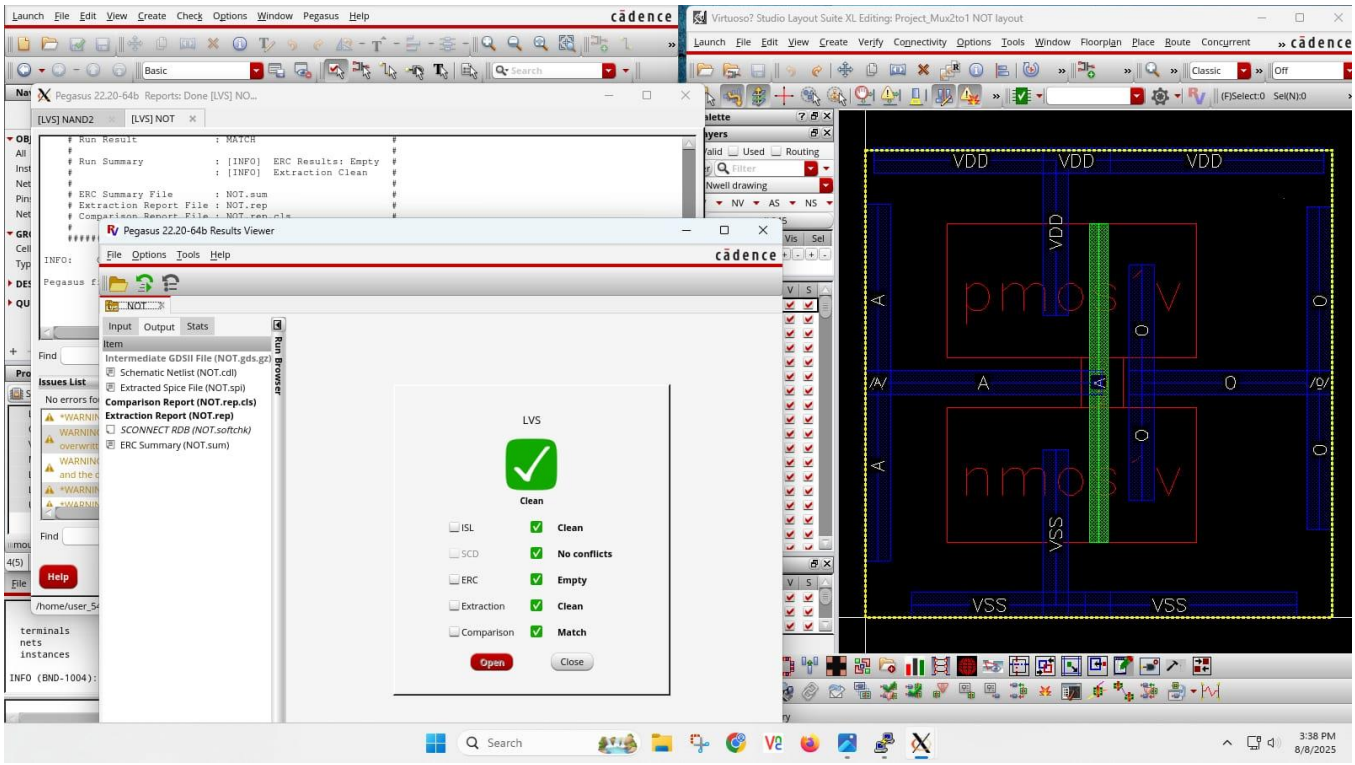
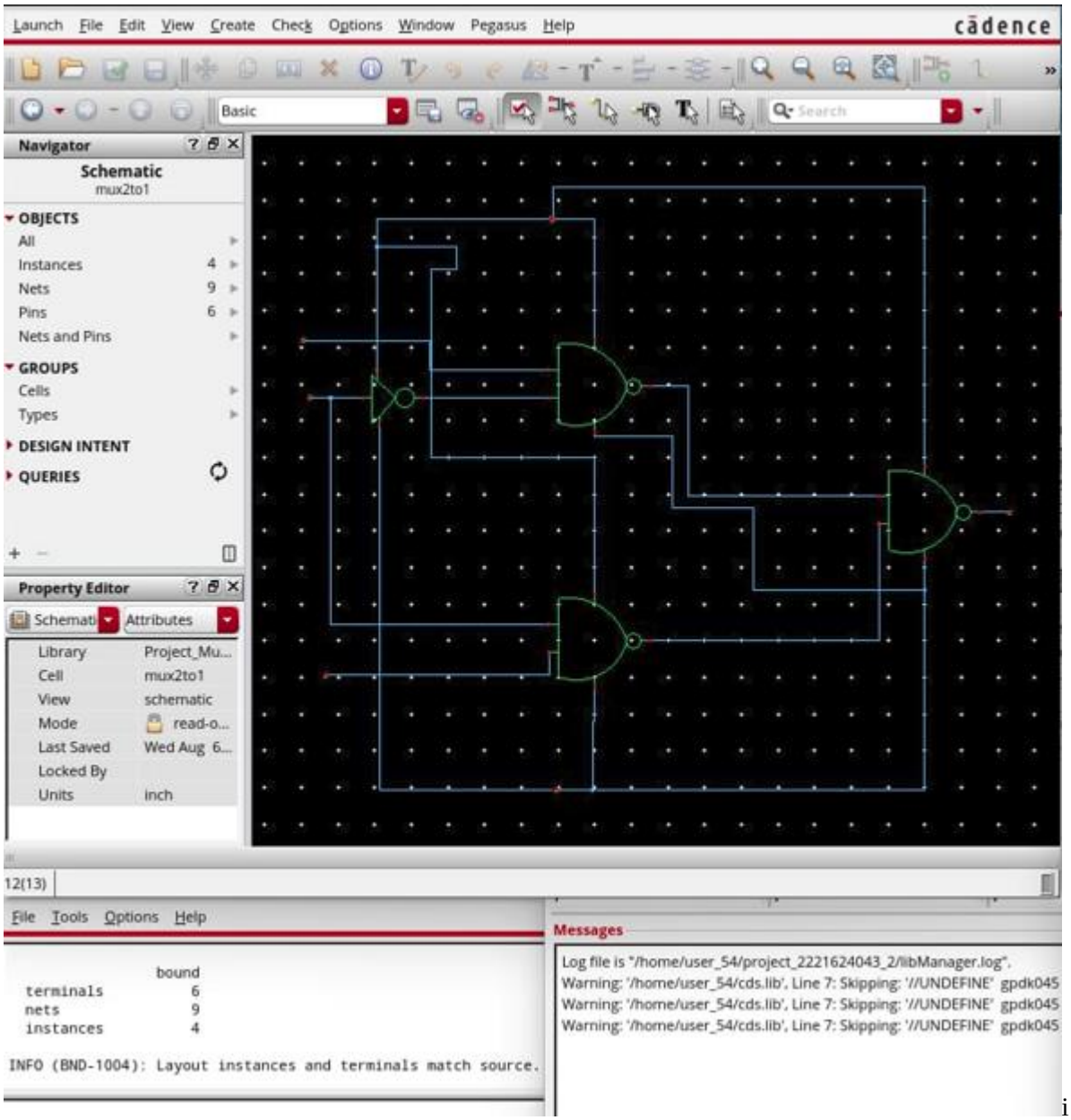


Figure 24: NOT LVS check



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Figure 25: MUX 2x1 Schematic

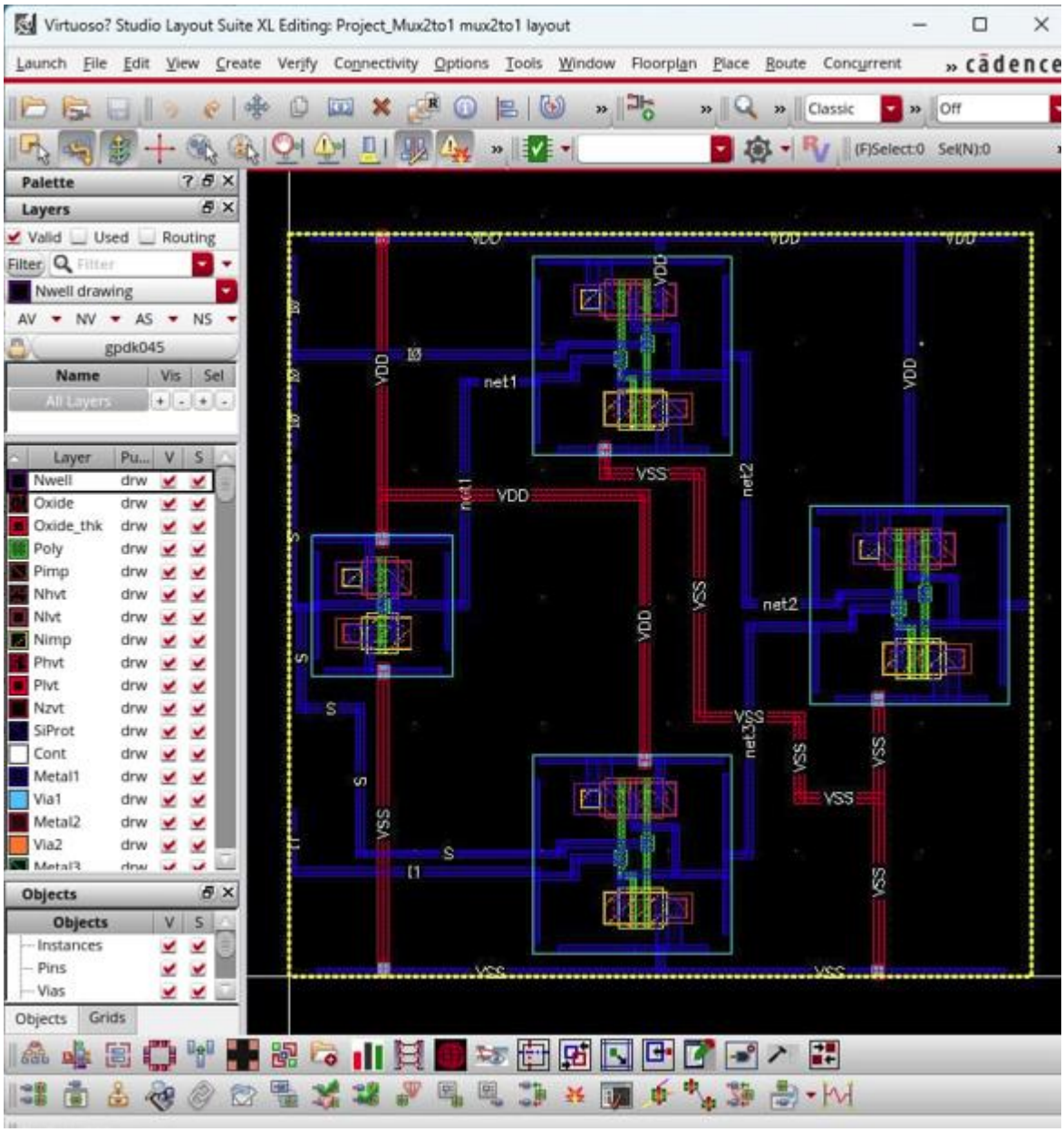


Figure 26: MUX 2x1 Layout

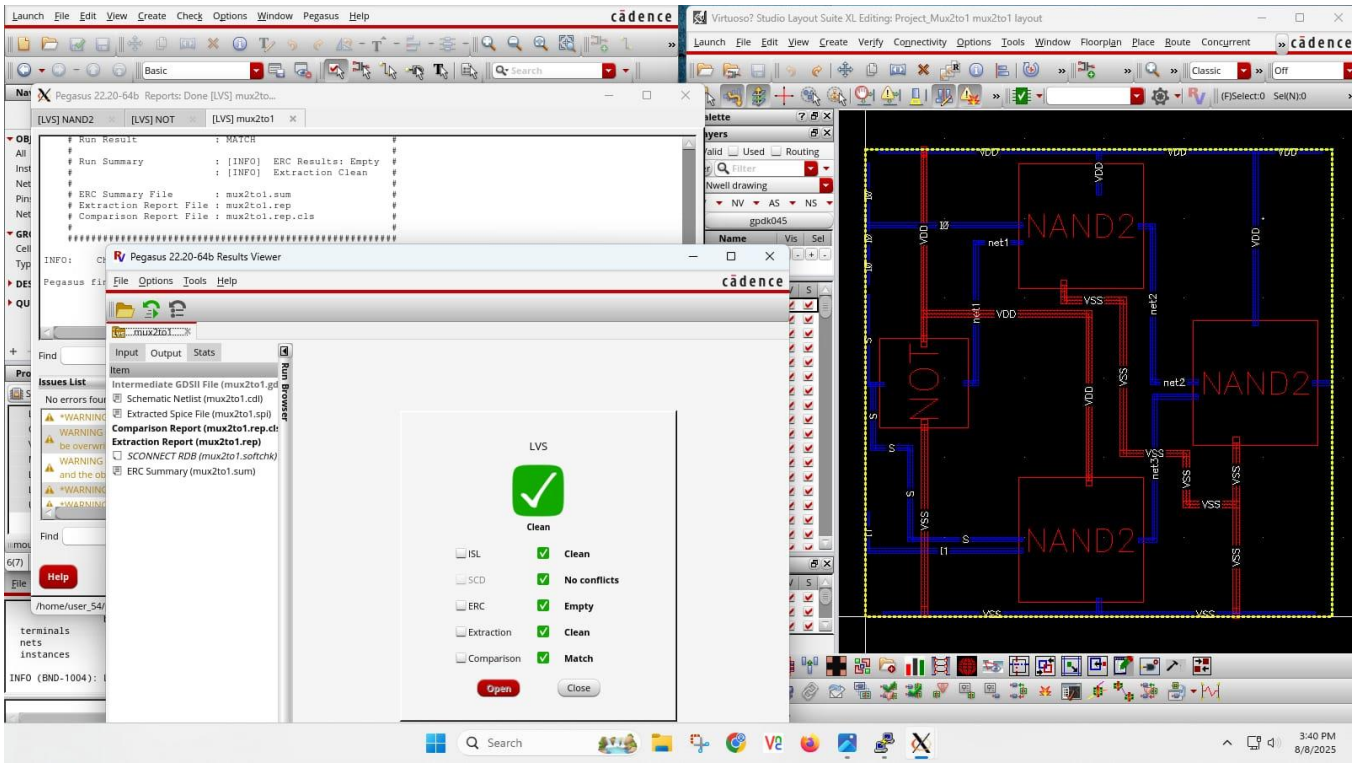


Figure 27: MUX 2x1 LVS check

NOR3

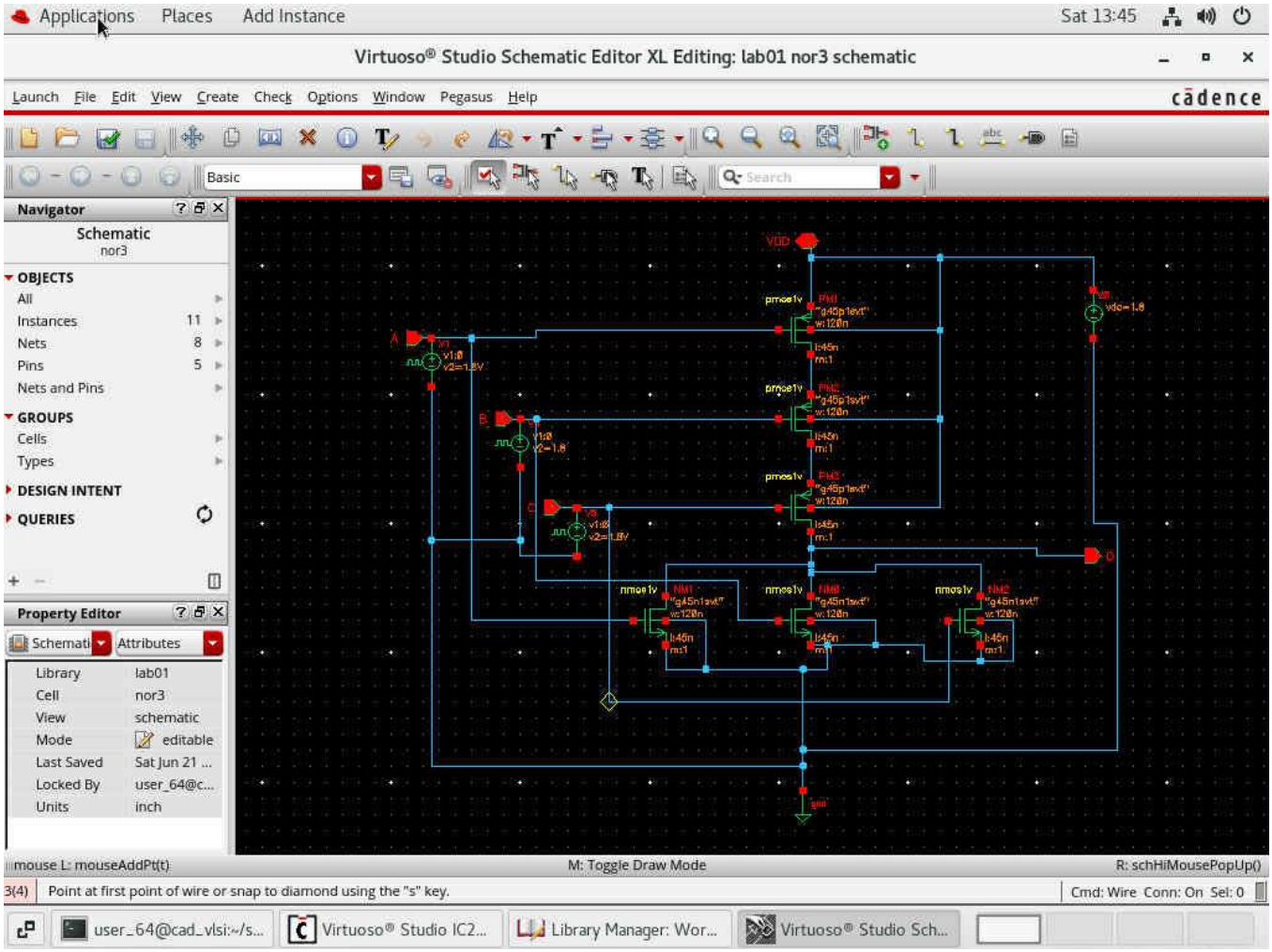


Figure 27: NOR3 Schematic

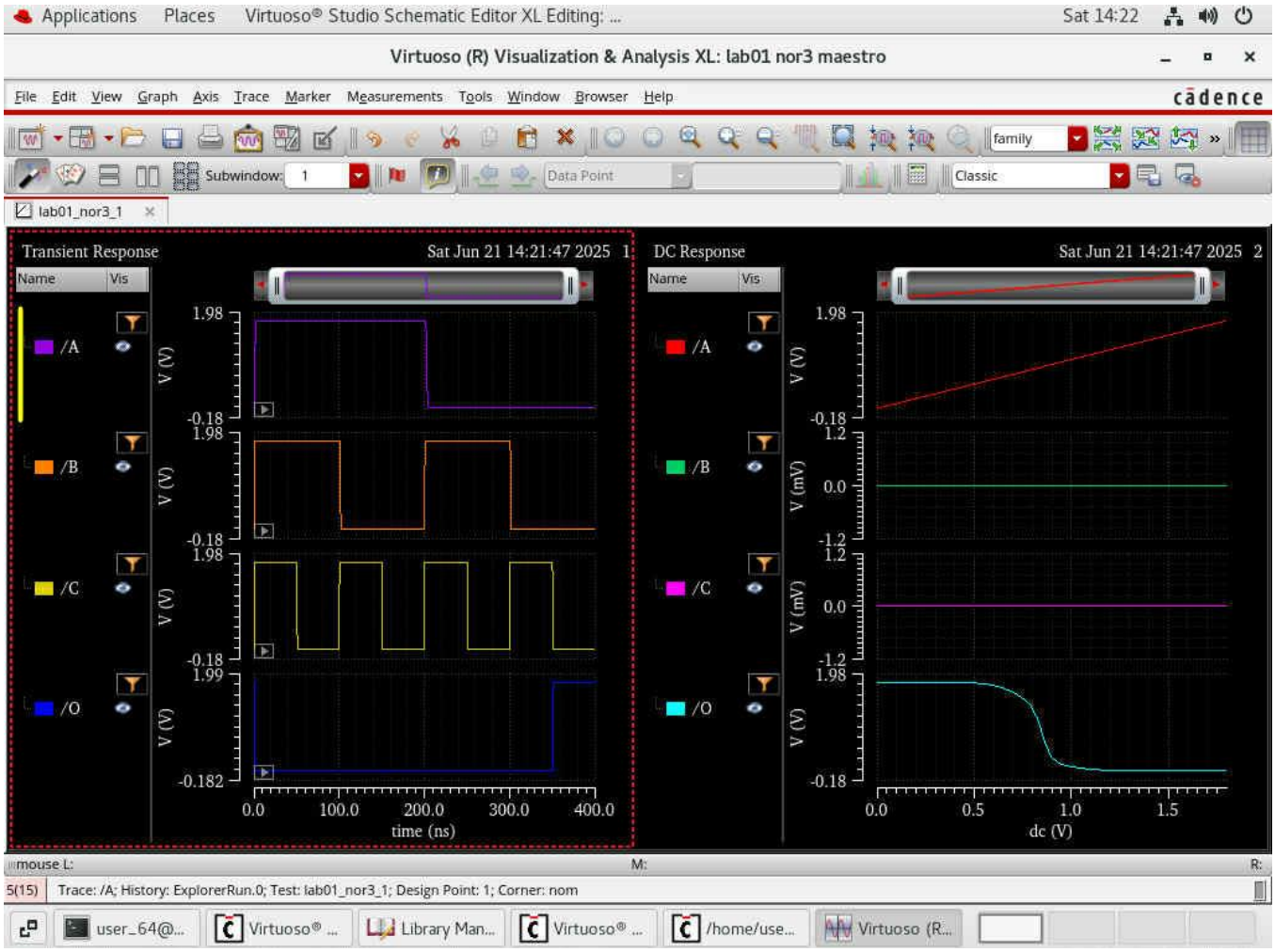


Figure 28: NOR3 Transient Analysis